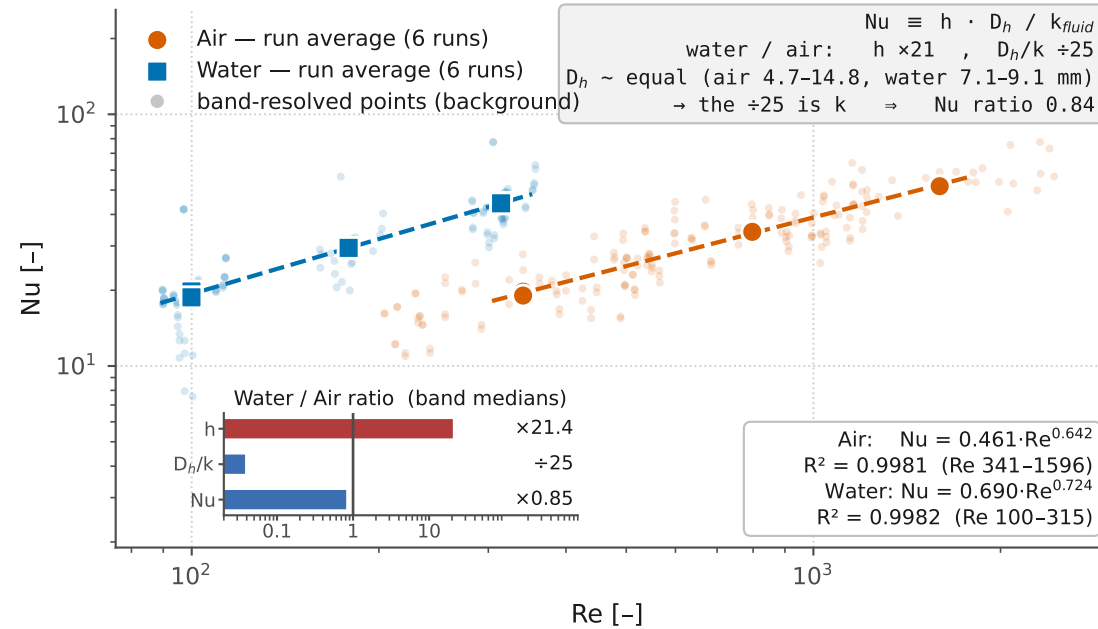
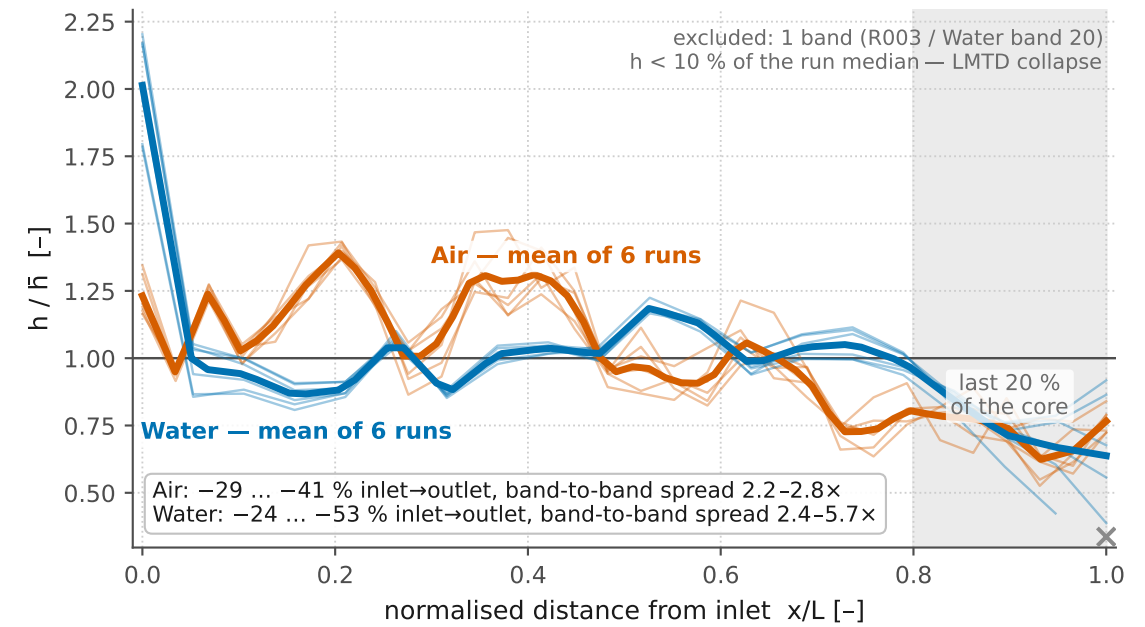


Band-resolved analysis of a TPMS gyroid heat exchanger: from six flow regimes to a design correlation set

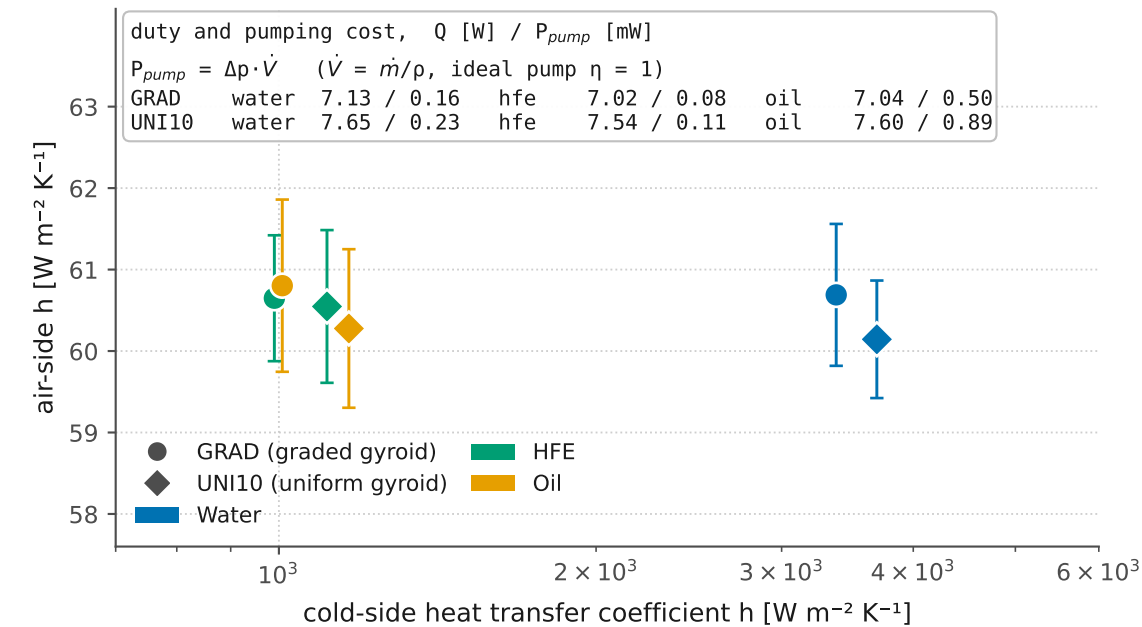
(a) Design correlation $Nu(Re)$



(b) The mean hides a 2-6× band-to-band variation

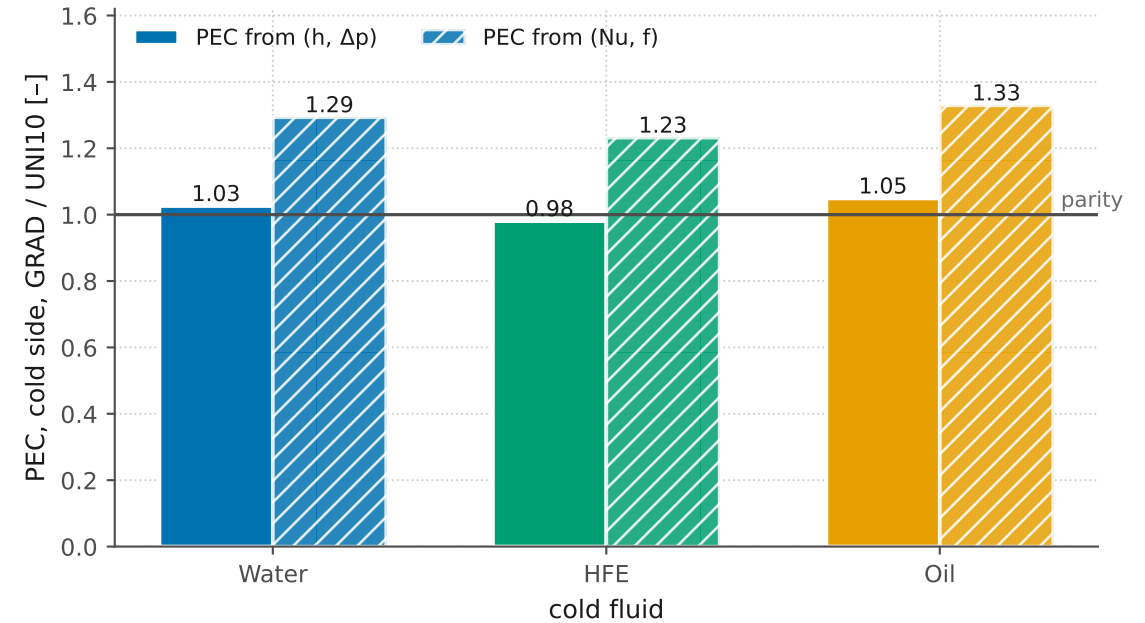


(c) Cold-fluid choice sets the pumping cost, not the duty



cold-side h varies $3.7\times \rightarrow$ duty stays at $Q = 7.02-7.65$ W;
 on UNI10 oil costs $3.9\times$ the pumping power of water for the same duty

(d) Verdict at equal pumping power — and the validation behind it



$PEC = (X_G/X_U)/(Y_G/Y_U)^{1/3}$, $X, Y = (h, \Delta p)$ or (Nu, f) , $G/U = GRAD/UNI10$; the $1/3$ exponent $\Rightarrow$ equal pumping power
 Validation (transient window 18-19 s, 50 time steps)
 energy balance $|Q_{air} - Q_{cold}|$: 0.33-0.67 % mean, $\leq 1.5\%$ worst step
 temporal scatter of h : air 1.2-1.7 %, cold liquid 0.02-1.14 %